

Exploration Of Application Migration to Cloud Environment

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Abstract—Cloud computing has strongly imposed itself on the area of research for both industry and academics requirements. Cloud providers are moving or planning to move their applications to the cloud environment attaining infrastructures and different cloud service organizations. This task includes a lot of details to be academic selected to tackle such a task raising up a lot of questions which needs to be addressed in a scientific and practical approach.

Most researches concerning this issue stressing on the challenges of the migration process to the cloud. But few studies stresses on the life running of practical applications in the cloud environment monitoring some selected performance measures to be compared before and after migration .The objective of this process could assist in taking a critical decision regarding either to keep or modify the practical migrated application.

The Objective of this research is to evaluate the performance measures of a real running transaction processing system (TPS) as a selected case study before and after migration to the cloud environment.

Plotting and analysis of the conducted results over the Amazon EC2 cloud environment as public cloud and a selected specific private cloud indicates 52% compared to the performance measures before migration which strongly support the usage of cloud computing approach for practical case studies without concerning about any performance degradation. The lower overhead cost for migration to public cloud environments proves the robustness of the proposed approach for this research.

Keywords—TPS, Cloud Computing, Performance measure, Application migration, Database applications.

I. INTRODUCTION

The term information system is an academic study of systems with a specific reference to information and the complementary networks of hardware and software that people and organizations use to collect, filter, process, create and also distribute data [1].

Transaction processing is a style of computing that divides work into individual, indivisible operations, called transactions. A transaction processing system (TPS) or transaction server is a software system, or software/hardware combination, that supports transaction processing.

Transaction processing system features such as Performance, Continuous availability, Data integrity, Ease of use, Modular growth [2].

The needed resources for TPS are Databases for transaction processing may be constructed using hierarchical, network, or relational structures, Suitable Infrastructure is needed for Databases according to its complexity levels from servers , networks , OS , clients PC's , smart phones , tablets, Storage , applications.

Resources for TPS can be provided on premise only till 2006 when Amazon.com introduced the Elastic Compute Cloud, A new paradigm was introduced from that date for deployment TPS systems over the internet which is called cloud computing [3].

The definition of cloud computing provided by the US National institute of standards and technology (NIST), “Cloud computing is a model for enabling convenient, on demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction”[4].

Cloud Computing Provides some advantages that makes it attractive to business owners such as No up-front investment as it uses a pay-as you-go pricing model so to run a TPS in the cloud you simply rents resources from the cloud according to its own needs and pay for the usage, Lowering operating cost as Resources in a cloud environment can be rapidly allocated and de-allocated on demand, Highly scalable where Infrastructure providers pool large amount of resources from data centers and make them easily accessible, Easy access as Services hosted in the cloud are generally web-based. Therefore, they are easily accessible through a variety of devices with Internet connections, Reducing business risks and maintenance expenses By outsourcing the service infrastructure to the clouds, a service provider shifts its business risks (such as hardware failures) to infrastructure providers[5].

There are three service models (i.e., Software as a Service, Platform as a Service, and Infrastructure as a Service) and several deployment models (e.g., public, private, and hybrid) for cloud computing.

Although cloud computing has shown considerable advantages, it brings some challenges such as Automated service provisioning, Server consolidation, Traffic management and analysis, Storage technologies and data management.

One of the most important challenges that need to be addressed is studying the performance of the existing on premise TPS applications after migration to cloud environment.

The studies were based on a ticketing system which is a Life TPS system working to issue tickets for vessels passengers, This system was selected as it needs high availability and stability and it is accessed by variety of stake holders (Accountants, Officers, IT staff), So a verification was needed that this kind of TPS systems with its demands are covered with the cloud.

The structure of this paper is organized as follows: section 2 presents the related work in this field, section 3 presents the proposed approach used in this paper, Section 4 explains the implementation, Section 5 presents the experimental results of our testing, section 6 presents the conclusion.

II. RELATED WORK

The popularization of the term Cloud Computing was in 2006 when Amazon.com introduced the Elastic Compute Cloud [3].

During this 8 years plenty of research has been made in this field but few has concentrated on migrating the existing enterprise systems to the cloud.

Mohagheghi and Sæther [6] have identified Some Software Engineering challenges related to migration of legacy application to the service cloud paradigm. This was a part of REMICS (Reuse and Migration of legacy application to Interoperable Cloud Services) project which was a research project that had been held from 2010 to 2013 to develop a set of model driven methods and tools to support in migrating legacy system to the service cloud paradigm.

Muhammad et. al. [7],[8] shared their observations of migrating an open source application to the cloud environment. Important quality attributes for migration such as portability of solutions, scalability, modularity, and consistent view of the system for users have been identified. Modifications have been made to the architecture of the open source solution to meet the new cloud environment quality attributes but no performance testing has been made to the application before and after the migration.

Khajeh-Hosseini et. al.[9] introduced The Cloud Adoption Toolkit toolkit which provides a framework for decision makers to help them in determining the feasibility and value of deployment application in the cloud environment.

Rongen [10] used this tool on an independent case study "an enterprise architecture collaboration and communication software package" to investigate the feasibility of deploying of this application to the cloud. Rongen evaluated this tool to be useful do a quick assessment of the requirements for the application.

Wei Hao et. al. [11] considered that service Oriented Architecture SOA is very important in cloud computing and service performance can be affected due to the widely distributed service providers, so they suggested to solve this problem by migrating services to different platforms considering the service selection problem and migration in cloud , they proposed a framework which facilitates service migration. They also designed a cost model with the decision algorithm to determine the right adjustment of service selection and migration..

Jyhjong Lin [12] presented a method for directing the migration of on-premise applications to selected clouds. This method employs design models as Business Process Management (BPM) lifecycle and The Open Group Architecture Framework (TOGAF) framework to support an effective migration of the on-premise applications into the clouds. For illustration, the method is applied to the migration of a CSS Customer Support System application to its cloud version.

Rashmi et. al. [13] discussed the challenges of migration to cloud from business factors and technical factors considerations. They proposed a five phased waterfall model for cloud migration based on iterative waterfall model from software development life cycle. They said that this model can enable organizations to guarantee that the migration cost wouldn't exceed its benefits.

III. THE PROPOSED APPROACH

a. General Idea

The proposed approach in this research is to test performance before and after migration to the cloud on a selected case study "ticketing system". Two groups of performance measures were measured. Group one was measured with Hp load runner.[14], group 2 was measured by oracle sql developer [15]. Four Scenarios were created which represents true actions on the case study, and loaded into the load runner and results were plotted for group 1. Queries were selected and its performance was tested with sql developer for group 2.

Scenario 1 is Admin Add Trip Action on the system, Scenario 2 is a Successful Issuing tickets transaction, Scenario 3 is Failure in Issuing tickets due to a validation raise, and Scenario 4 is Admin Update Trip Action.

b. Selected Case Study

The case study is enterprise ticketing system which is a Transaction Processing System (TPS) developed by Oracle tools "Oracle forms and Oracle Reports 10g Release 2"[16]

and written in oracle PL/SQL. The ticketing system is used for reservation and issuing tickets on vessels that carry passengers, cargo between different countries. The system was migrated to Apex 4.2[17] to be cloud based application that can be deployed on a private and public cloud platform.

Figure 1 and 2 shows UML design diagrams of the ticketing system

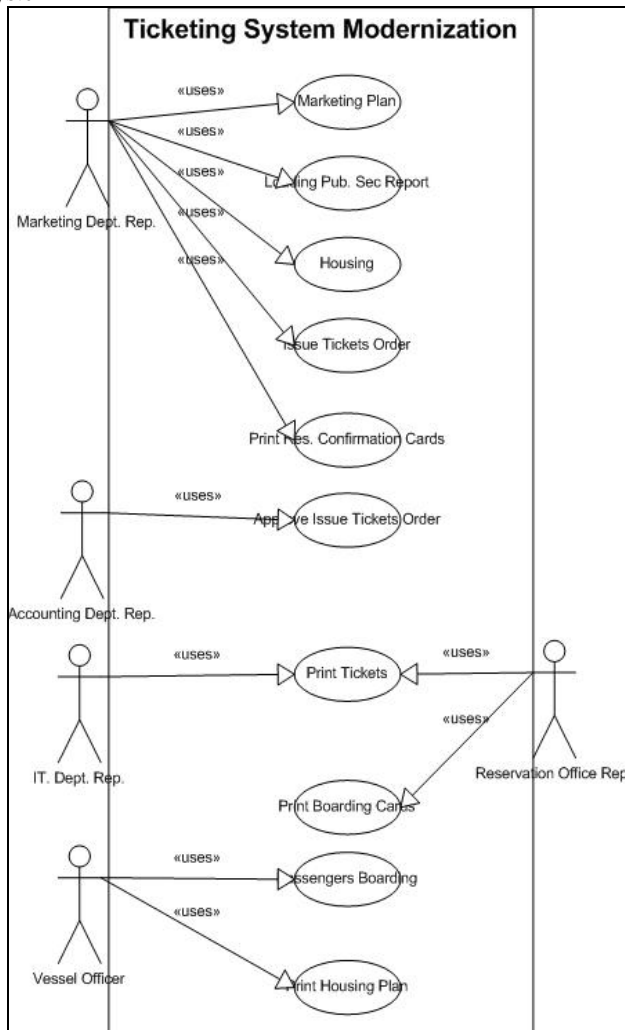


FIGURE 1: USE CASE DIAGRAM OF THE TICKETING SYSTEM

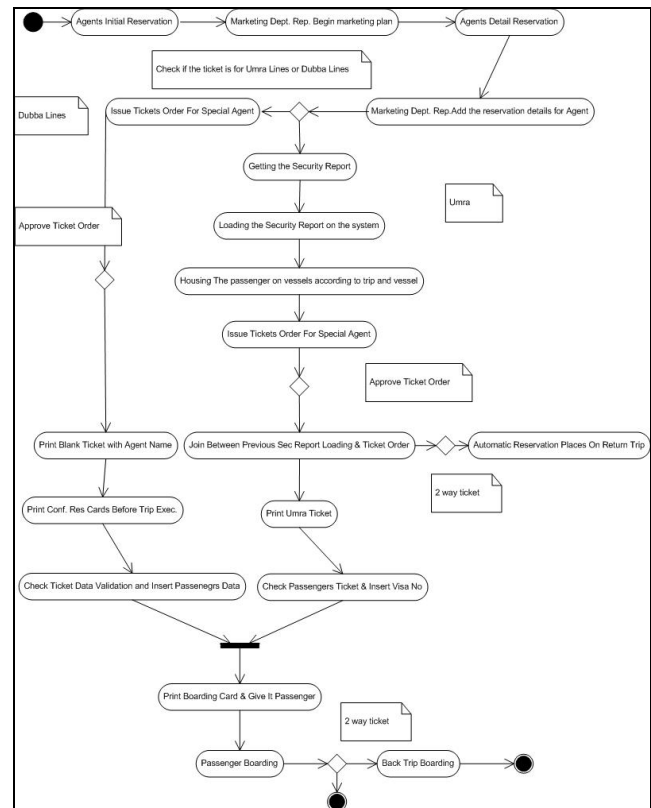


FIGURE 2: ACTIVITY DIAGRAM OF THE TICKETING SYSTEM

c. Performance measures used

Two groups of performance measures are measured,

Group 1:

1. Total Throughput (bytes): Displays the amount of data received from the server.
2. Total Hits: Displays the number of requests sent to the server during the load test.
3. Average Transaction Response Time: Displays the average time to perform transactions during each second of the load test.
4. Average Throughput (bytes / second): Displays the amount of data received from the server at any given second.
5. Average Hits per second: Displays the number of requests per second sent to the server during the load test.

Group 2:

1. CPU used by this session: is the amount of CPU time (in 10s of milliseconds) used by a session between when a user call started and ended
2. DB time: is the amount of elapsed time (in microseconds) spent performing Database user-level calls.
3. Bytes received via SQL*Net from client: is the total number of bytes received from the client over SQL*Net.
4. Bytes sent via SQL*Net to client: is the total number of bytes sent to the client from the foreground processes

d. Servers Configuration

1. On premise server configuration: Intel Zeon Processor 2.5 GHz, 4 GB RAM, Windows 2003 R2 Enterprise.
2. Private Cloud configuration: Intel Zeon Processor 2.5 GHz (2 processors) , 16 GB RAM, Windows 2008 R2 Standard 64-bit
3. Public Cloud configuration on Amazon EC2: Intel Zeon Processor 2.5 GHz, 1 GB RAM, Windows 2008 R2 Standard 64-bit.

IV. IMPLEMENTATION

As mentioned earlier in section III, the Hp load runner was used to perform performance testing on the selected case study. For Group 1 ,the scenarios were created by Load Runner Virtual User Generator. Then Scenarios are loaded in the load runner controller to perform the performance testing on the loaded scenario. Results of Performance are testing are displayed in the load runner Analysis.

V. Experimental Results

Table 1 summarizes the obtained results for on premise performance measures testing.

Table 2 summarizes the obtained results for private cloud performance measures testing.

Table 3 summarizes the obtained results for public cloud performance measures testing.

From the obtained results, Figure 3 shows that sc#4,sc#1 on private cloud has the highest total throughput "amount of data received to the user from server 60,437,499 bytes while Sc#2 and Sc#3 on private and public cloud has almost equal figures.

Figure 4 shows that Sc#2 On premise has the highest total hits 45,854 then Sc#3 On Premise comes after with 24,717 at Figure 5 shows that Sc#1 on public cloud has least Average response time 0.45 seconds , while Sc#2 on premise has highest Average response time with 8.597 seconds. then comes Sc#3 on premise scoring 6.816 seconds.

TABLE (1) EXPERIMENTAL RESULTS ON PREMISE

SC#	TOTAL THROUGHPUT	TOTAL HITS	AVG RESPONSE TIME	AVG THROU GHPUT	AVG HITS PER SECOND
Sc #1	31,723,101	6,756	2.732	77,563	16.518
Sc #2	40,894,608	45,854	8.597	98,304	110.26
Sc #3	48,521,068	24,717	6.816	121	61.638
Sc #4	30,574,243	4,622	2.399	73,673	11.137

TABLE (2) EXPERIMENTAL RESULTS ON PRIVATE CLOUD

SC#	TOTAL THROUGHPUT	TOTAL HITS	AVG RESPONSE TIME	AVG THROU GHPUT	AVG HITS PER SECOND
Sc #1	56,892,981	3,950	5.218	134,82	9.36
Sc #2	43,588,990	3,675	3.998	77,423	6.528
Sc #3	45,366,762	3,630	3.314	83,395	6.673
Sc #4	60,437,499	3,779	4.896	146,34	9.15

TABLE (3) EXPERIMENTAL RESULTS ON PUBLIC CLOUD

SC#	TOTAL THROUGHPUT	TOTAL HITS	AVG RESPONSE TIME	AVG THROU GHPUT	AVG HITS PER SECOND
Sc #1	5,423,462	617	0.45	13,491	1.535
Sc #2	49,820,304	3,886	4.245	88,334	6.89
Sc #3	46,233,471	1,811	1.517	63,682	2.494
Sc #4	49,318,029	1,469	2.144	106,29	3.166

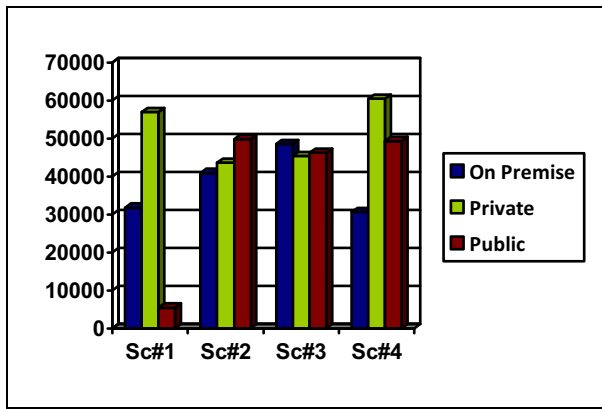


FIGURE 3: TOTAL THROUGHPUT

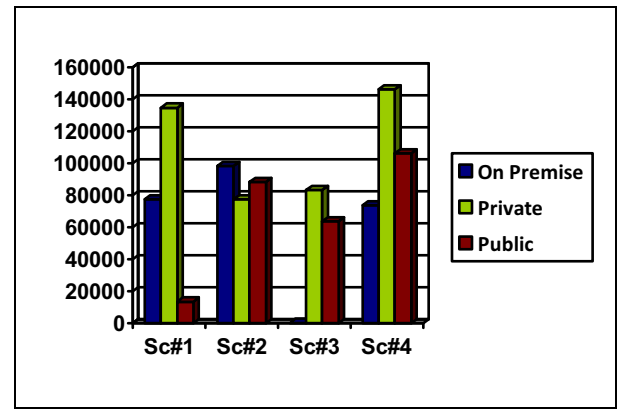


FIGURE 6: AVERAGE THROUGHPUT

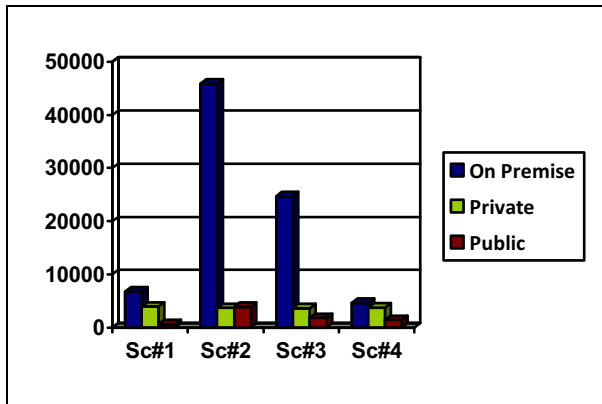


FIGURE 4: TOTAL HITS

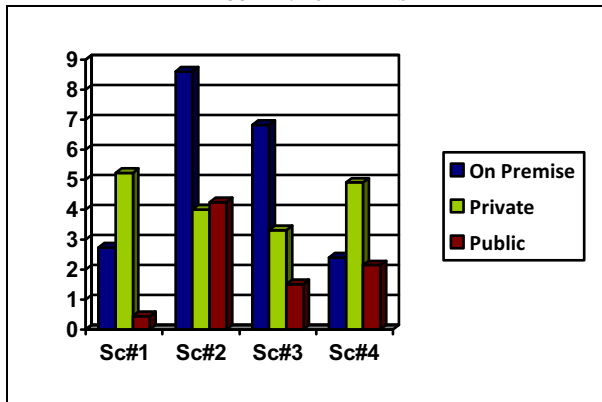


FIGURE 5: AVERAGE RESPONSE TIME

VI. CONCLUSION

In this research, performance measures of a practical TPS application is analyzed in its on premise state and after migration to cloud environment including a public and private base. This include deep focus on the UML diagrams for the TPS selected case study.

The measured performance parameters on Amazon EC2 environment, are compared to its counterpart on premise and private cloud environments.

Four scenarios are designed based on the applicability of TPS selected case study: Admin Add Trip Action on the system, Successful Issuing tickets transaction, Failure Issuing tickets due to a validation raise and Admin Update Trip Action. This ensures the wide scope of coverage applicability of the case study on premise, on private or public cloud environment.

The proposed research includes the measurements of selected performance measures such as: total throughput, total hits, average transaction response time, average throughput and average hits per second.

The performance measures results shows that a public cloud, with its sophisticated services superimposes the performance on a private cloud and on premise.

It should be noticed that a careful study of overhead cost for migration to private cloud environment.

Also, it is eligible to study other public cloud providers to generalize the proposed claim of this research regarding both the performance measures and overhead cost.

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